

University of Gujrat

Department of Software Engineering

Database Systems (Lab)

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Table Of Contents

Non-Normalized Method....3
1st Normalized Form....4
2nd Normalized Form....5
3rd Normalized Form....6

Non-Normalized Form

As we can see, that data is not normalized in this table. So, we will normalize this data in 3 Steps.

Student_ID	Student_Name	Program	Program_Director	Enrolled Courses(Course_Code, Course_Name, Grade)		
101	Alice Smith	Computer Science	Dr. Turing	(CS10, Database, A), (CS11, Programming, B)		
102	Bob Jones	Business	Dr. Yunus	(MGT1, Management, B), (CS10, Database, C)		
103	Charlie Brown	Computer Science	Dr. Turing	(CS11, Programming, A)		

1st Normalized Form

We have converted the table to its 1st Normalized form. We have used flattening to change this table to 1NF. This is why we broke down Course Code, Course Name and Grade into Separate columns.

Student_ID	Student_Name	Program	Program_Director	Course_Code	Course_Name	Grade
101	Alice Smith	Computer Science	Dr. Turing	CS10	Database	A
101	Alice Smith	Computer Science	Dr. Turing	CS11	Programming	B
102	Bob Jones	Business	Dr. Yunus	MGT1	Management	B
102	Bob Jones	Business	Dr. Yunus	CS10	Database	C
103	Charlie Brown	Computer Science	Dr. Turing	CS11	Programming	A

2nd Normalized Form

Since, 2NF involves removing partial dependencies which happens when non-key attributes are dependent on only one of the composite keys. Therefore We identify these and break down it into tables.

Students

Student_ID	Student_Name	Program	Program_Director
101	Alice Smith	Computer Science	Dr. Turing
101	Alice Smith	Computer Science	Dr. Turing
102	Bob Jones	Business	Dr. Yunus
102	Bob Jones	Business	Dr. Yunus
103	Charlie Brown	Computer Science	Dr. Turing

Courses

Course_Code	Course_Name
CS10	Database
CS11	Programming
MGT1	Management
CS10	Database
CS11	Programming

Grades

Student_ID	Course_Code	Grade
101	CS10	A
101	CS11	B
102	MGT11	B
102	CS10	C
103	CS11	A

3rd Normalized Form

In 3NF we make sure that the tables must be in 2NF and have no transitive dependencies. Transitive dependencies is where a non-key attribute depends on another non-key attribute. So, we further break

Students

Student_ID	Student_Name	Program
101	Alice Smith	Computer Science
101	Alice Smith	Computer Science
102	Bob Jones	Business
102	Bob Jones	Business
103	Charlie Brown	Computer Science

Courses

Course_Code	Course_Name
CS10	Database
CS11	Programming
MGT1	Management
CS10	Database
CS11	Programming

Programs

Program	Program_Director
Computer Science	Dr. Turing
Computer Science	Dr. Turing
Business	Dr. Yunus
Business	Dr. Yunus
Computer Science	Dr. Turing

3rd Normalized Form

Grades

		Student_ID	Course_Code	Grade		
		101	CS10	A		
		101	CS11	B		
		102	MGT11	B		
		102	CS10	C		
		103	CS11	A		